

BASEL-MULHOUSE-FREIBURG AP, France

WMO: 072990

Lat: 47.6144N

Lon: 7.5100E

Elev: 270

StdP: 98.13

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.6	-6.0	-11.4	1.5	-7.1	-8.7	1.8	-4.4	12.2	10.7	10.5	8.8	1.8	330	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	32.1	20.9	30.0	20.2	28.1	19.5	21.8	29.6	21.0	28.2	20.2	26.8	2.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.5	24.5	18.6	13.8	23.6	17.9	13.3	22.7	64.8	29.8	61.8	28.3	59.0	27.0	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.2	6.1	DB	-12.4	35.2	3.9	1.9	-15.2	36.6	-17.5	37.7	-19.6	38.8	-22.4	40.2	(4)
			WB	-12.8	23.9	3.8	0.9	-15.5	24.5	-17.7	25.1	-19.9	25.6	-22.7	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	2.2	3.3	6.8	10.5	14.8	18.4	20.1	19.7	15.5	11.5	6.1	2.9
	DBStd	7.42	4.38	4.40	3.84	3.75	3.50	3.58	3.19	3.27	3.18	3.77	3.78	4.15
	HDD10.0	961	244	188	111	40	4	0	0	0	1	29	123	221
	HDD18.3	2872	501	421	356	237	121	43	18	22	96	213	366	478
	CDD10.0	1336	1	1	13	54	151	252	313	302	166	74	8	1
	CDD18.3	204	0	0	0	0	9	46	73	65	11	1	0	0
	CDH23.3	2198	0	0	0	22	139	491	771	652	115	8	0	0
CDH26.7	746	0	0	0	1	27	161	288	248	21	1	0	0	
Wind	WSAvg	2.6	3.0	2.9	2.9	2.7	2.7	2.4	2.4	2.2	2.2	2.3	2.5	2.9
Precipitation	PrecAvg	775	49	51	50	60	79	89	74	103	54	47	71	46
	PrecMax	1069	95	153	102	117	130	184	135	181	152	106	178	84
	PrecMin	519	6	10	9	16	41	11	43	28	9	13	11	6
	PrecStd	132	24	32	22	28	25	39	24	43	34	26	40	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	16.1	21.1	26.0	29.6	33.6	34.9	35.3	29.2	24.3	17.7	13.7
		MCWB	10.6	9.8	12.1	15.1	18.9	21.4	21.5	21.2	19.1	16.8	13.1	10.5
	2%	DB	12.2	13.7	18.6	23.0	26.8	30.7	32.1	31.8	26.5	21.6	15.0	12.0
		MCWB	9.3	8.8	11.0	13.7	17.1	20.6	20.9	20.8	18.8	16.0	11.7	9.7
	5%	DB	10.1	11.7	15.8	20.5	24.6	28.4	29.9	29.4	24.2	19.3	13.1	10.3
		MCWB	7.8	8.1	10.2	12.9	16.3	19.9	20.2	20.1	17.6	15.0	10.5	8.3
	10%	DB	8.2	9.6	13.5	18.1	22.2	26.2	27.7	27.1	21.9	17.2	11.5	8.7
		MCWB	6.4	6.9	9.3	11.9	15.4	18.8	19.5	19.7	16.6	14.0	9.5	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.7	13.1	15.9	20.0	23.2	22.8	22.9	20.6	17.8	13.6	11.2
		MCDB	13.6	14.7	18.7	23.6	27.2	30.4	31.5	30.6	26.6	22.3	16.6	13.1
	2%	WB	9.5	9.4	11.7	14.5	18.5	21.5	21.7	21.7	19.4	16.6	12.1	9.8
		MCDB	11.8	13.0	16.5	21.3	25.0	28.6	29.9	29.3	25.2	20.4	14.5	11.7
	5%	WB	7.9	8.3	10.7	13.4	17.1	20.4	20.8	20.8	18.2	15.4	10.9	8.6
		MCDB	9.8	11.2	14.9	19.4	22.6	27.1	28.3	27.7	22.8	18.7	12.9	10.1
	10%	WB	6.5	7.0	9.6	12.3	16.0	19.3	19.9	19.9	17.1	14.3	9.7	7.2
		MCDB	8.1	9.5	13.4	17.4	20.9	25.1	26.5	26.3	21.2	16.9	11.3	8.6
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.8	10.0	11.2	10.8	11.4	11.8	11.6	11.0	9.3	6.8	5.7
		MCDBR	8.2	11.5	14.5	16.0	15.6	15.4	16.1	16.1	14.8	12.4	9.4	7.4
		MCWBR	5.9	7.2	8.3	8.1	7.3	6.9	6.4	6.7	7.6	7.1	6.2	5.4
	5% WB	MCDBR	7.8	10.3	12.4	14.3	13.5	14.2	14.5	14.2	13.0	11.3	8.7	6.9
		MCWBR	5.9	6.8	7.5	7.5	6.7	6.6	6.2	6.3	6.9	6.7	6.0	5.3
Clear-Sky Solar Irradiance	taub	0.311	0.336	0.370	0.407	0.416	0.421	0.420	0.413	0.390	0.367	0.332	0.308	
	taud	2.444	2.368	2.308	2.225	2.246	2.274	2.271	2.304	2.348	2.396	2.445	2.446	
	Ebn at Noon	771	821	844	844	849	845	840	829	814	773	734	726	
	Edh at Noon	66	88	109	130	133	130	128	119	104	85	66	59	
All-Sky Solar Radiation	RadAvg	1.06	1.86	3.10	4.33	5.05	5.95	5.66	4.92	3.72	2.32	1.22	0.85	
	RadStd	0.14	0.30	0.48	0.65	0.51	0.54	0.51	0.47	0.36	0.21	0.14	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (112 neighbors)	N/A	N/A	N/A	+0.75	+0.39	+0.29	N/A	N/A	+52	+14

Nomenclature: See separate page